Abstract algebra homework6

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Problem 1

Let R be a commutative ring and consider the polynomial ring $R[x]$.

(a) $\deg(f+g) \leq \max\{\deg(f), \deg(g)\}$

Let $f, g \in R[x] \setminus \{0\}$ with $\deg(f) = n$ and $\deg(g) = m$. Write

$$f(x) = \sum_{i=0}^{n} a_i x^i, \qquad g(x) = \sum_{i=0}^{m} b_i x^i,$$

with $a_n \neq 0$ and $b_m \neq 0$. Without loss of generality assume $n \geq m$ (the argument is symmetric otherwise). Extend the coefficients of g by setting $b_i = 0$ for $m < i \leq n$. Then

$$f(x) + g(x) = \sum_{i=0}^{n} (a_i + b_i) x^i.$$

This is a polynomial in which every coefficient of x^i for $i > n = \max\{n, m\}$ is zero (there are no terms of that degree in either f or g). Hence either $f + g = 0$, in which case $\deg(f+g) = -\infty \leq \max\{n, m\}$ by convention, or $f + g \neq 0$ and its degree is at most $n = \max\{\deg f, \deg g\}$. In either case,

$$\deg(f+g) \leq \max\{\deg(f), \deg(g)\}. \qquad \blacksquare$$

(b) If R is an integral domain, $\deg(fg) = \deg(f) + \deg(g)$

Let $f, g \in R[x] \setminus \{0\}$ with $\deg(f) = n$, $\deg(g) = m$, leading coefficients $a_n \neq 0$, $b_m \neq 0$ as above. Then

$$f(x)g(x) = \sum_{i=0}^{n} \sum_{j=0}^{m} a_i b_j x^{i+j}.$$

The coefficient of x^{n+m} in this product is exactly $a_n b_m$ (every other pairing (i, j) with $i+j = n+m$, $i \leq n$, $j \leq m$ forces $i = n, j = m$). Since R is an integral domain and $a_n, b_m \neq 0$, we have $a_n b_m \neq 0$. Also every coefficient of x^k for $k > n + m$ vanishes, since there is no way to write $k = i + j$ with $i \leq n, j \leq m$. Hence $fg \neq 0$ and

$$\deg(fg) = n + m = \deg(f) + \deg(g).$$

Consequence: R is an integral domain $\iff$ $R[x]$ is an integral domain.

($\Rightarrow$) Suppose R is an integral domain. If $f, g \in R[x]$ are both nonzero, then by the computation above $fg \neq 0$ (its degree is the finite number $\deg f + \deg g$, so in particular $fg \neq 0$). Also $R[x]$ is commutative with $1 \neq 0$ since R is. Hence $R[x]$ is an integral domain.

($\Leftarrow$) Suppose $R[x]$ is an integral domain. The map $R \hookrightarrow R[x]$ sending $r \mapsto r$ (the constant polynomial) is an injective ring homomorphism, identifying R with a subring of $R[x]$ containing 1. A subring (containing 1) of an integral domain is itself an integral domain: it is commutative, has $1 \neq 0$, and has no zero divisors since it inherits this property from $R[x]$. Hence R is an integral domain. ■

(c) Units of $R[x]$ when R is an integral domain

Claim. The units of $R[x]$ are precisely the constant polynomials a with $a \in R^\times$ (a unit of R).

Proof. ($\Leftarrow$) If $a \in R^\times$, let $a^{-1} \in R$ be its inverse in R. Viewing a, a^{-1} as constant polynomials, $a \cdot a^{-1} = 1$ in $R[x]$, so a is a unit of $R[x]$.

($\Rightarrow$) Suppose $f \in R[x]$ is a unit, so there exists $g \in R[x]$ with $fg = 1$. In particular $f, g \neq 0$ (else $fg = 0 \neq 1$, using $1 \neq 0$ in R). Since R is an integral domain, part (b) gives

$$\deg(f) + \deg(g) = \deg(fg) = \deg(1) = 0.$$

Since $\deg(f), \deg(g) \geq 0$ are nonnegative integers summing to 0, we must have $\deg(f) = \deg(g) = 0$. Thus $f = a$ is a nonzero constant polynomial, and $g = b$ is a nonzero constant with $ab = 1$ in R, i.e. $a \in R^\times$. ■

Problem 2

Let R be an integral domain of characteristic p, a prime, and define $\varphi : R \to R$ by $\varphi(x) := x^p$.

(a) φ is a ring homomorphism (the Frobenius endomorphism)

Multiplicativity. Since R is commutative, for all $x, y \in R$,

$$\varphi(xy) = (xy)^p = x^p y^p = \varphi(x)\varphi(y).$$

Identity. $\varphi(1) = 1^p = 1$.

Additivity. By the binomial theorem (valid in any commutative ring),

$$(x+y)^p = \sum_{k=0}^{p} \binom{p}{k} x^k y^{p-k}.$$

For $0 < k < p$, the binomial coefficient $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ is divisible by p: indeed $p \mid p!$ but $p \nmid k!$ and $p \nmid (p-k)!$ since $k, p-k < p$ and p is prime, so the factor of p in $p!$ survives in $\binom{p}{k}$. Write $\binom{p}{k} = p\, m_k$ for some integer m_k. Since R has characteristic p, we have $p \cdot 1_R = 0$, hence $p \cdot r = 0$ for every $r \in R$ (as $p \cdot r = (p \cdot 1_R) \cdot r$). Therefore

$$\binom{p}{k} x^k y^{p-k} = (p\, m_k)\, x^k y^{p-k} = m_k \cdot (p \cdot x^k y^{p-k}) = 0 \quad \text{for } 0 < k < p.$$

All the middle terms of the binomial expansion vanish, leaving

$$(x+y)^p = x^p + y^p, \qquad \text{i.e.} \qquad \varphi(x+y) = \varphi(x) + \varphi(y).$$

Hence φ is a ring homomorphism. ■

(b) If R is finite, φ is an automorphism

Since $\varphi : R \to R$ is a ring homomorphism (part (a)), it suffices to show φ is injective; injectivity of a map from a finite set to itself of the same (finite) cardinality automatically implies surjectivity (pigeonhole principle), so φ would be bijective, hence an automorphism.

Injectivity. The kernel of φ (as an additive group homomorphism) is $\{x \in R : x^p = 0\}$. Suppose $x \in \ker \varphi$, so $x^p = 0$, i.e. $\underbrace{x \cdot x \cdots x}_{p \text{ times}} = 0$. Since R is an integral domain (no zero divisors), a product of nonzero elements is nonzero; by induction, if $x \neq 0$ then $x^p \neq 0$. Contrapositively, $x^p = 0 \implies x = 0$. Hence $\ker \varphi = \{0\}$, so φ is injective.

Since R is finite and $\varphi : R \to R$ is an injective function, φ is also surjective (an injective self-map of a finite set is a bijection). Therefore φ is a bijective ring homomorphism, i.e. an automorphism of R. ■

Problem 3

Let R be a commutative ring.

(a) An intersection of ideals is an ideal

Let $\mathcal{I}$ be a family of ideals of R, and let $J := \bigcap_{I \in \mathcal{I}} I$. We verify the ideal axioms.

J is a subgroup of $(R, +)$. Every $I \in \mathcal{I}$ contains 0 (as 0 is in every ideal), so $0 \in J$. If $a, b \in J$, then $a, b \in I$ for every $I \in \mathcal{I}$; since each I is closed under subtraction, $a - b \in I$ for every $I \in \mathcal{I}$, hence $a - b \in J$. So J is a subgroup of $(R, +)$.

J absorbs multiplication by R. Let $a \in J$ and $r \in R$. For every $I \in \mathcal{I}$, $a \in I$, and since I is an ideal, $ra \in I$. As this holds for every $I \in \mathcal{I}$, we get $ra \in J$. Hence $J = \bigcap_{I \in \mathcal{I}} I$ satisfies all the ideal axioms and is an ideal of R. ■

(b) (X) is the smallest ideal of R containing X

Recall $\mathcal{I}(X) = \{I \subset R : I \text{ is an ideal and } I \supset X\}$ and $(X) = \bigcap_{I \in \mathcal{I}(X)} I$. **$\mathcal{I}(X)$ is nonempty.** R itself is an ideal of R and $R \supset X$, so $R \in \mathcal{I}(X)$. This guarantees the intersection defining (X) is over a nonempty family, so (X) is well-defined. **(X) an ideal.** $\mathcal{I}(X)$ is a family of ideals of R, so by part (a), $(X) = \bigcap_{I \in \mathcal{I}(X)} I$ is an ideal of R. **$X \subseteq (X)$.** Every $I \in \mathcal{I}(X)$ satisfies $I \supset X$ by definition of $\mathcal{I}(X)$. Since X is contained in every member of the family being intersected,

$$X \subseteq \bigcap_{I \in \mathcal{I}(X)} I = (X).$$

Minimality. Let J be any ideal of R with $X \subseteq J$. Then J itself satisfies the defining condition of $\mathcal{I}(X)$ (it is an ideal containing X), so $J \in \mathcal{I}(X)$. Since (X) is the intersection of *all* members of $\mathcal{I}(X)$, and J is one such member,

$$(X) = \bigcap_{I \in \mathcal{I}(X)} I \subseteq J.$$

Combining the three boxed facts: (X) is an ideal containing X, and it is contained in every ideal containing X. This is precisely the statement that (X) is the smallest ideal of R containing X. ■

(c) Explicit description of (X)

Define

$$S := \left\{ \sum_{i=1}^{n} r_i a_i \ : \ n \in \mathbb{N}^+, \ r_i \in R, \ a_i \in X \right\}.$$

We show $S = (X)$ by proving $S \in \mathcal{I}(X)$ (which gives $(X) \subseteq S$) and $S \subseteq I$ for every $I \in \mathcal{I}(X)$ (which gives $S \subseteq (X)$).

Step 1: S is an ideal of R containing X.

Contains X: for $a \in X$, take $n = 1$, $r_1 = 1$: then $a = 1 \cdot a \in S$. So $X \subseteq S$; in particular $S \neq \varnothing$.

Closed under addition: if $u = \sum_{i=1}^{n} r_i a_i$ and $v = \sum_{j=1}^{m} s_j b_j$ are two elements of S (with $a_i, b_j \in X$), then

$$u + v = \sum_{i=1}^{n} r_i a_i + \sum_{j=1}^{m} s_j b_j$$

is again a finite R-linear combination of elements of X (with $n + m$ terms), so $u + v \in S$.

Closed under negation (hence subtraction, given closure under addition): $-u = \sum_{i=1}^{n} (-r_i) a_i \in S$.

Absorbs multiplication by R: for $r \in R$,

$$r \cdot u = r \sum_{i=1}^{n} r_i a_i = \sum_{i=1}^{n} (r r_i) a_i \in S.$$

Hence S is an ideal of R with $S \supset X$, i.e. $S \in \mathcal{I}(X)$.

Step 2: $(X) \subseteq S$.

Since $S \in \mathcal{I}(X)$ and (X) is the intersection of *all* ideals in $\mathcal{I}(X)$, in particular $(X) \subseteq S$.

Step 3: $S \subseteq (X)$.

Let $I \in \mathcal{I}(X)$ be arbitrary, and let $u = \sum_{i=1}^{n} r_i a_i \in S$ with $a_i \in X$, $r_i \in R$. Since $X \subseteq I$, each $a_i \in I$; since I is an ideal, $r_i a_i \in I$ for each i; since I is closed under addition, $u = \sum_i r_i a_i \in I$. As $I \in \mathcal{I}(X)$ was arbitrary, u lies in every ideal of $\mathcal{I}(X)$, so $u \in \bigcap_{I \in \mathcal{I}(X)} I = (X)$. Hence $S \subseteq (X)$.

Combining Steps 2 and 3, $S = (X)$, i.e.

$$(X) = \left\{ \sum_{i=1}^{n} r_i a_i \ : \ n \in \mathbb{N}^+, \ r_i \in R, \ a_i \in X \right\}. \qquad \blacksquare$$

(Remark: if $X = \varnothing$, the formula is interpreted via the empty sum as $(X) = \{0\}$, consistent with (X) being the intersection of all *ideals of R, which always contains 0 and equals $\{0\}$ precisely when the zero ideal itself is included in the intersection.)*

Problem 4

Let R be an integral domain. Show $a \in R$ is irreducible $\iff$ (a) is maximal among proper principal ideals of R.

Throughout, recall two standard facts we use freely:

- $u \in R$ is a unit $\iff$ $(u) = R$: if u is a unit, $u^{-1} u = 1 \in (u)$, so $(u) = R$; conversely if $(u) = R$ then $1 = ru$ for some r, so u is a unit.

- (Problem 5) a, b are associates ($a = ub$ for a unit u) $\iff$ $(a) = (b)$.

By definition, a is *irreducible* if $a \neq 0$, a is not a unit, and whenever $a = bc$ for $b, c \in R$, either b or c is a unit.

($\Rightarrow$) Irreducible implies (a) is maximal among proper principal ideals

Suppose a is irreducible. Since a is not a unit, $(a) \neq R$, so (a) is a proper ideal.

Let (b) be any proper principal ideal with $(a) \subseteq (b)$; we must show $(a) = (b)$. Since $a \in (a) \subseteq (b)$, we can write $a = bc$ for some $c \in R$. As a is irreducible, either b or c is a unit.

If b were a unit, then $(b) = R$, contradicting that (b) is proper. Hence c is a unit. Then $b = ac^{-1}$, so a and b are associates ($a = bc$ with c a unit, equivalently $b = ac^{-1}$ with c^{-1} a unit). By Problem 5, $(a) = (b)$.

So the only proper principal ideal containing (a) is (a) itself: (a) is maximal among proper principal ideals of R.

($\Leftarrow$) (a) maximal among proper principal ideals implies a irreducible

Suppose (a) is a proper principal ideal, maximal among proper principal ideals of R. In particular a is not a unit (else $(a) = R$ would not be proper), and $a \neq 0$ (the zero ideal is not maximal among proper principal ideals in an integral domain that is not a field, and if R were a field there are no irreducibles or proper principal ideals to speak of besides (0); we take $a \neq 0$ as given, consistent with the standing hypothesis for irreducibility).

Suppose $a = bc$ for some $b, c \in R$; we must show b or c is a unit. We have $a \in (b)$, so $(a) \subseteq (b)$.

Case 1: $(b) = R$. Then b is a unit, and we are done.

Case 2: $(b) \neq R$, i.e. (b) is a proper (principal) ideal containing (a). By maximality of (a) among proper principal ideals, $(a) = (b)$. By Problem 5, a and b are associates, so $b = au$ for some unit $u \in R^\times$. Substituting into $a = bc$:

$$a = bc = (au)c = a(uc).$$

So $a(1 - uc) = 0$. Since $a \neq 0$ and R is an integral domain, $1 - uc = 0$, i.e. $uc = 1$. Hence $c = u^{-1}$ is a unit.

In both cases, b or c is a unit. Since a is also nonzero and not a unit, a is irreducible. ■

Problem 5

Let R be an integral domain and $a, b \in R$. Show a, b are associates $\iff$ $(a) = (b)$.

Recall a, b are *associates* if $a = ub$ for some unit $u \in R^\times$ (equivalently $b = u^{-1}a$).

($\Rightarrow$)

Suppose $a = ub$ for a unit u. Then $a \in (b)$, so $(a) \subseteq (b)$. Also $b = u^{-1}a$, so $b \in (a)$, giving $(b) \subseteq (a)$. Hence $(a) = (b)$.

($\Leftarrow$)

Suppose $(a) = (b)$.

If $a = 0$: since $a \in (a) = (b)$... more directly, $(a) = (0)$ forces $(b) = (0)$ as well, and since $b = 1 \cdot b \in (b)$, this means $b = 0$. So $a = b = 0$, and $0 = 1 \cdot 0$ with 1 a unit, so a, b are (trivially) associates. Now assume $a, b \neq 0$.

Since $a \in (a) = (b)$, write $a = rb$ for some $r \in R$. Since $b \in (b) = (a)$, write $b = sa$ for some $s \in R$. Substituting,

$$a = rb = r(sa) = (rs)a \implies a(1 - rs) = 0.$$

Since $a \neq 0$ and R is an integral domain, $1 - rs = 0$, i.e. $rs = 1$. Hence r (and s) is a unit of R. Since $a = rb$ with r a unit, a and b are associates. ■

Problem 6

Compute the monic gcd of

$$f(x) = 2x^3 + x^2 + x - 1, \qquad g(x) = 4x^3 - 2x^2 - 6x + 3$$

in $\mathbb{Q}[x]$ via the Euclidean algorithm.

Step 1: Divide g by f

$$g - 2f = (4x^3 - 2x^2 - 6x + 3) - (4x^3 + 2x^2 + 2x - 2) = -4x^2 - 8x + 5.$$

So

$$g = 2f + r_1, \qquad r_1(x) = -4x^2 - 8x + 5.$$

Step 2: Divide f by r_1

Dividing $f = 2x^3 + x^2 + x - 1$ by $r_1 = -4x^2 - 8x + 5$:

$$f = \left(-\tfrac{1}{2}x + \tfrac{3}{4}\right) r_1 + r_2, \qquad r_2(x) = \tfrac{19}{2}x - \tfrac{19}{4} = \tfrac{19}{4}(2x - 1).$$

(Check: $\left(-\frac{1}{2}x + \frac{3}{4}\right)(-4x^2 - 8x + 5) = 2x^3 + 4x^2 - \frac{5}{2}x - 3x^2 - 6x + \frac{15}{4} = 2x^3 + x^2 - \frac{17}{2}x + \frac{15}{4}$, and f minus this is $\left(x + \frac{17}{2}x\right) + \left(-1 - \frac{15}{4}\right) = \frac{19}{2}x - \frac{19}{4}$, as claimed.)

Since nonzero constant multiples do not affect divisibility (constants are units in $\mathbb{Q}[x]$), we may replace r_2 by the associate

$$\tilde{r}_2(x) := 2x - 1$$

and continue the algorithm with $\tilde{r}_2$ in place of r_2.

Step 3: Divide r_1 by $\tilde{r}_2$

$$r_1 = -4x^2 - 8x + 5 = (-2x - 5)(2x - 1) + 0.$$

Indeed: $(-2x - 5)(2x - 1) = -4x^2 + 2x - 10x + 5 = -4x^2 - 8x + 5 = r_1$ exactly, with remainder 0.

Conclusion

The last nonzero remainder is $\tilde{r}_2(x) = 2x - 1$, so

$$\gcd(f, g) \doteq 2x - 1 \quad \text{(up to units)},$$

and the **unique monic gcd** is

$$\boxed{\gcd(f, g) = x - \tfrac{1}{2}.}$$

Verification. One can check directly that f and g both factor as

$$f(x) = (2x - 1)(x^2 + x + 1), \qquad g(x) = (2x - 1)(2x^2 - 3),$$

(both verified by direct expansion), and $x^2 + x + 1$ (discriminant $-3 < 0$, no real roots) shares no common factor with $2x^2 - 3$ (roots $\pm\sqrt{3/2}$, irrational) over $\mathbb{Q}$. Hence $2x - 1$ is indeed the gcd up to units, confirming $x - \frac{1}{2}$ is the monic gcd.